Evaluating functions

1 Consider the function $y = 4x - 3$.

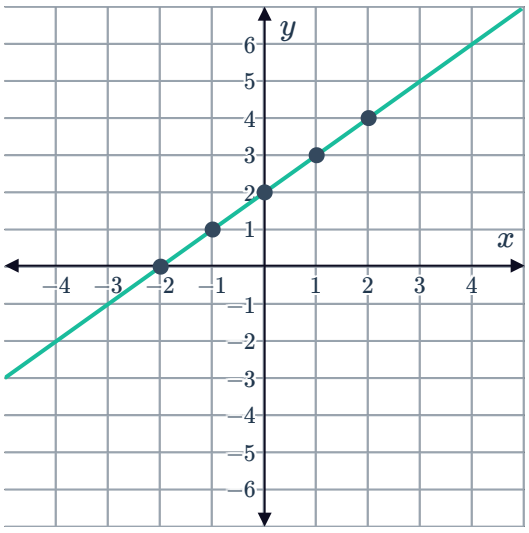
a Identify the independent variable of the function.

b Identify the dependent variable of the function.

2 For a function f, state what $f\,(3)$ represents.

3 Suppose that $(7, -6)$ is an ordered pair that satisfies the function g. Write this situation using function notation.

4 Consider the following graph of the function $f\,(x)$.



a Find $f\,(0)$.

b Find $f\,(-2)$.

c Find the value of x such that $f\,(x) = 3$

5 Consider the function $f\,(x) = 8x + 6$.
Determine the output produced by the input value $x = 5$.

6 Consider the function $f\,(x) = -9x^2 - 8x - 4$.
Determine the output produced by the input value $x = -5$.

7 If $f\,(x) = 4x + 4$, find:

a $f\,(2)$.

b $f\,(-5)$.

8 If $f\,(x) = 3x - 1$, find:

a $f\,(3)$

b $f\,(-4)$

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9 If $g(x) = \dfrac{7x}{4}$, find:

a $g(5)$ **b** $g(-4)$

10 If $f(x) = \dfrac{2x-3}{5}$, find:

a $f(4)$ **b** $f(-5)$

11 Consider the function $f(x) = 2x^2 - 2x + 5$.
Evaluate $f\left(\dfrac{1}{2}\right)$.

12 Consider the function $f(x) = x^2 + 8x$.

a Form an expression for $f(a)$. **b** Form an expression for $f(b)$.

13 Consider the function $p(x) = x^2 + 8$.

a Evaluate $p(2)$. **b** Form an expression for $p(m)$.

14 Consider each of the following equations.

i Rewrite the equation using function notation $f(x)$.

ii Find the value of $f(3)$.

a $x + 3y = 6$ **b** $-6x + 5y = 7$

c $y + 6x^2 = 3 - x$

15 Consider the equation $x - 4y = 8$.
Assume that y is a function of x.

a Rewrite the equation using function notation $f(x)$.

b Find the value of $f(12)$.

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16 Consider each of the following equations.

i Rewrite the equation using function notation $f(x)$.

ii Find the value of $f(2)$.

a $5x - 3y = 7$

b $y - 4x^2 = 5 + x$

17 Find the value of x that satisfies the equation $\frac{1}{4} = \frac{x + 12}{8}$.

18 Consider the function $f(x) = -4x^2 - 5x + 6$.
Determine $f(-5)$.

19 State whether each of the following is true or false.

a The cost of a pizza depends on the size of the pizza, so the cost of a pizza is a function of the size of the pizza.

b The size of a pizza depends on the price of the pizza, so the size of a pizza is a function of the price of a pizza.

20 The model $P(t) = 3500 \times 2^t$ describes the population of fish in a lake after t years, where $t = 0$ is the initial time when the population of fish was first recorded.
Determine and interpret the correct value of$P(2)$.

Additional questions

21 Sharon works at a pizza shop and makes $\$15$ per hour. Her wages can be represented by the function $W(t) = 15t$, where t represents the number of hours that she works.

a Explain what the expression $W(19)$ represents.

b Explain what the equation $W(n) = 165$ represents.

c Find her wages after working 26 hours.

22 Mindy can model her tomato plant's height in inches over time using the function $h(t) = 2^t$, where t is the number of weeks after Mindy planted it.

a Explain what the equation $h(4) = 16$ represents.

b Find the height of Mindy's tomato plant after 6 weeks.

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23 Consider the function $f(x) = ax + b$. If $f(4) = 7$ and $f(-1) = -3$, find the values of a and b. Explain how you arrive at the answer.

24 Describe a scenario that can be represented by the function $f(x) = \dfrac{120}{x} + 5$.

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